Behavioral Finance and Technical Analysis

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Chapter Roadmap

9

9.1 **The Behavioral Critique**

Why investors may not be rational, and why mispricing can survive even when some investors are

9.2 **Technical Analysis and Behavioral Finance**

Trends, machine learning, sentiment indicators, and the warning that goes with all of them

Where this chapter sits

Chapter 8 found return patterns that a simple risk model cannot explain, and three ways to read them: a missing risk premium, genuine inefficiency, or data mining. Behavioral finance takes the second reading seriously and asks what kind of investor would produce those patterns.

It is not a rival theory of prices so much as a different account of where prices come from — and, as the last part of 9.1 argues, that is both its appeal and its weakness.

Key Concepts and Skills

1	**Name the errors**	Distinguish the information-processing mistakes — forecasting errors, overconfidence, conservatism, representativeness — from the behavioral biases built on top of them.
2	**Use prospect theory**	Explain how a value function defined over gains and losses, rather than wealth, changes what an investor treats as risky.
3	**Explain why mispricing persists**	Set out the three limits to arbitrage and show why a rational investor may be unable to correct a price that is wrong.
4	**Read the law-of-one-price evidence**	Describe the twin-share, carve-out, and closed-end fund cases and say why they are the cleanest evidence available.
5	**Assess technical analysis**	Describe trend and sentiment indicators, and explain why apparent patterns in price data are so easy to find and so hard to trust.

9.1

The Behavioral Critique

- Information processing and behavioral biases
- Prospect theory
- Limits to arbitrage and the law of one price
- Bubbles, and how the critique holds up

The Critique Stands on Two Legs

9.1

Both are needed. Either one alone leaves prices correct.

1 **Investors do not always process information correctly**

→ **Irrationality**

They misread evidence, weight recent events too heavily, and are more confident in their own judgement than the record justifies. So they arrive at the wrong estimate of value.

2 **Arbitrage cannot always correct the error**

→ **Limits to arbitrage**

Even an investor who sees the mispricing may be unable or unwilling to trade against it hard enough or long enough to remove it.

Irrationality by itself is not enough.
If rational investors could trade freely against the mistakes of others, prices would be correct however foolish the crowd — the foolish would simply lose money to the rational.

This is why the second leg carries as much weight as the first, and why the limits-to-arbitrage evidence later in this section matters more than any single catalogue of biases.

Information Processing: Four Errors

9.1

Error	What it is	What it produces in prices
Forecasting errors	Too much weight on **recent experience**, and forecasts that are too extreme given how uncertain the information really is	Prices that overreact to recent news — a high P/E on a firm that has just reported well
Overconfidence	Investors overestimate the precision of their own beliefs and abilities	Too much trading, too little diversification, and a preference for active management
Conservatism	Beliefs are updated too slowly when new evidence arrives	Prices that underreact — the drift after an earnings announcement
Representativeness	A small sample is treated as if it described the whole population	A short run of good results is read as a trend, producing extrapolation and later reversal

Note that the four do not point the same way: conservatism makes prices move too little, representativeness makes them move too much.

Underreaction and Overreaction, Explained

9.1

The two awkward findings from Chapter 8, each matched to an error.

Underreaction ← Conservatism

If investors update too slowly, the price moves only part of the way when news arrives and keeps moving afterwards as belief catches up. That is underreaction, and it looks like momentum over a few months.

Overreaction ← Representativeness

If a few good quarters are read as proof of a new era, the price is pushed past what the evidence supports.
That is overreaction, and it is corrected over the following years — the long-horizon reversal.

This is the strongest thing behavioral finance has to offer: one framework that produces both patterns, at the horizons where each is actually observed, from two well-documented features of human judgement.

Behavioral Biases

9.1

Even with correct beliefs, decisions can be inconsistent.

Framing

- The same choice is accepted or rejected depending on how it is described.
- “Made money in 8 of 10 years” versus “lost money in 2 of 10 years”

Mental accounting

- Money is kept in separate mental pots.
- Spending dividends freely but refusing to sell shares for the same amount

Regret avoidance

- Losses hurt more when the decision was unconventional.
- Losing 20% in a blue chip is forgiven; losing 20% in a small-cap is not

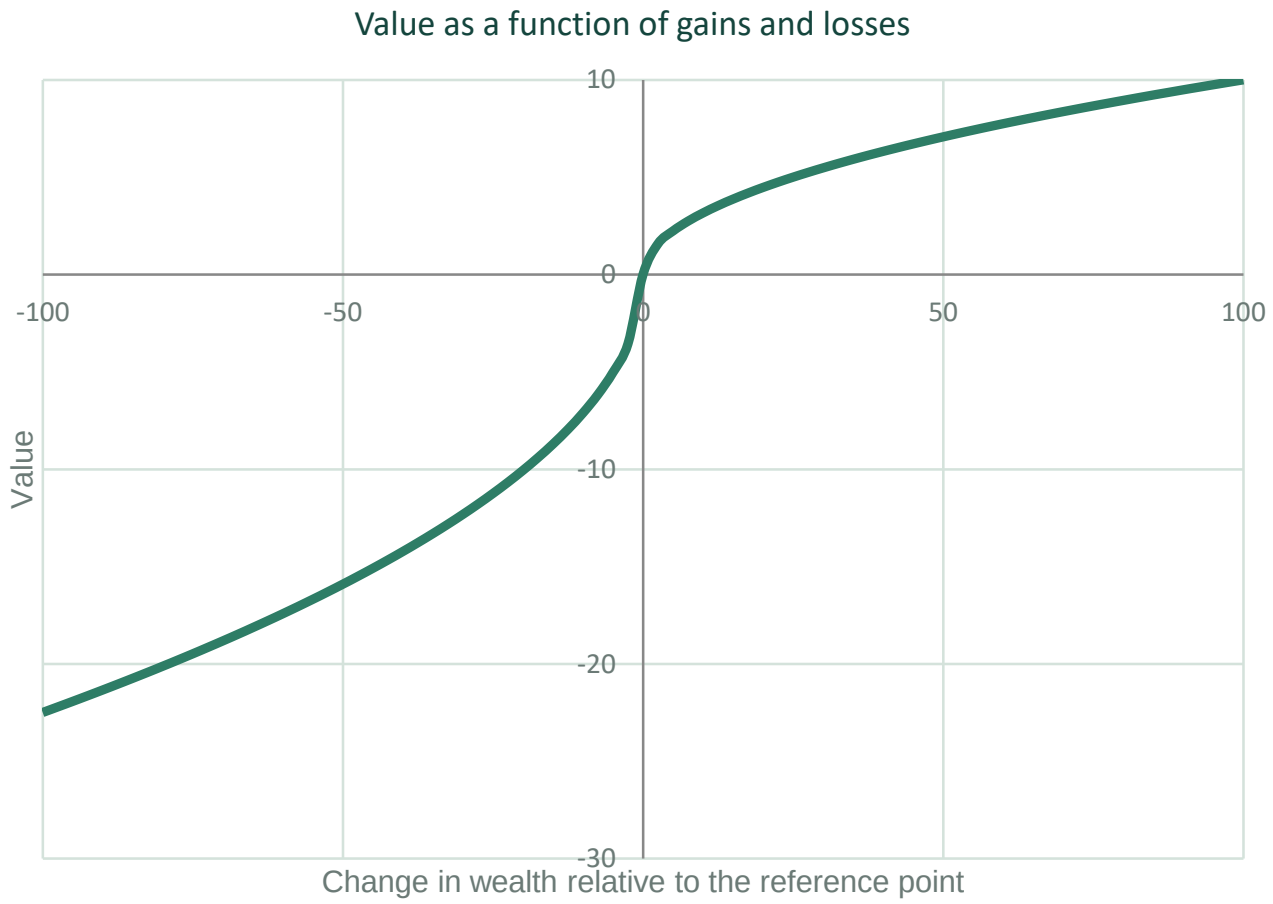
Affect

- Investors are drawn to stocks they feel good about — admired firms, familiar brands, the employer — independently of what those stocks are worth.
- Holding your employer’s stock and your favourite brand without extra analysis

Prospect Theory

9.1

Utility defined over gains and losses, not over the level of wealth.



Three features that matter

- Value is measured from a reference point — usually the purchase price or current wealth — not from total wealth.
- The curve is concave in gains and convex in losses, so an investor is risk-averse over gains and risk-seeking over losses.
- The loss side is steeper. A loss of a given size hurts roughly twice as much as the same gain pleases.
- Consequence: an investor who is sitting on a loss will take risks to get back to even that the same investor would refuse from a standing start.

Schematic, drawn with the standard parameterization (curvature 0.88, loss aversion 2.25).

Limits to Arbitrage

9.1

You are convinced a stock trading at **$80** is worth **$50**. The trade is obvious: short it. Here is what can go wrong — and none of it requires you to be wrong about the $50.

Fundamental risk
The mispricing can widen before it closes.

The price can go to $100 before it goes to $50. You are right about value and still lose money — and you may be forced to close the position before you are proved right.

Implementation costs
The trade may be expensive, or impossible to put on.

To short the stock you must first borrow it. There may be none to borrow, the fee may be large, and the lender can recall the shares at any time — forcing you to close at the worst moment. **Correcting a price that is too low only takes buying; correcting a price that is too high takes shorting. So overpricing survives longer than underpricing.**

Model risk
You may be the one who is wrong.

Perhaps $80 is right and your $50 is wrong. From the price alone you cannot tell "the market is mispricing this" from "my valuation is mistaken" — and that doubt limits how much you are willing to bet.

Notice when each of these bites hardest: when the price has moved furthest from value. That is exactly when your position is largest, your losses deepest, the borrow scarcest, and the investors who gave you the money are asking for it back. Arbitrage is weakest precisely where it is most needed.

Violations of the Law of One Price

9.1

Case 1 — Siamese twins : two share classes, one set of cash flows

The structure

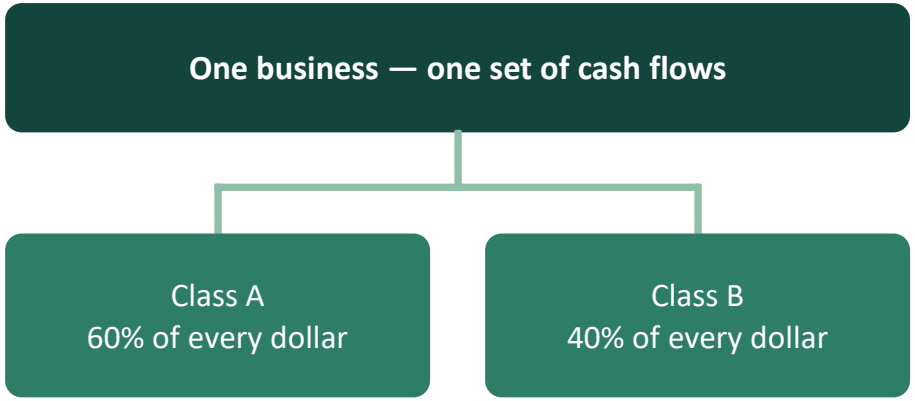
In 1907 two oil companies merged their operations but kept their shares separate.
By charter, every dollar the combined business earns is split 60% to one share class and 40% to the other. Nothing else distinguishes them.

What must be true

The total market value of the 60% class must be exactly 1.5 times the total market value of the 40% class.
Always, by arithmetic — no model, no forecast, no judgment about the oil price.

What the market did

The ratio wandered. The 60% class traded as much as 35% below parity, and at times 15% above — and the gap persisted for years, not days.
The two shares did not trade at parity until 2001. (**Royal Dutch Petroleum and Shell Transport & Trading**.)



By charter, not by valuation.

A Company Worth Less Than What It Owns

9.1

Case 2 — **Equity carve-outs**, *March 2000*

The structure. A technology firm owned a handheld-computer subsidiary. In March 2000 it sold 5% of that subsidiary to the public and announced that the rest would be handed to its own shareholders — about **1.5 subsidiary shares for every parent share**.

Where 1.5 comes from. Not a chosen premium — a division. The parent held about 503 million subsidiary shares and had about 330 million of its own shares outstanding: 503 ÷ 330 ≈ 1.5 per share.

What must be true. Parent price ≥ 1.5 × subsidiary price — because the parent holds those shares *plus* a profitable business *plus* cash.

Day one — 2 March 2000	
Subsidiary closing price	$95.06
Subsidiary inside one parent share (× 1.5)	$142.59
Parent closing price	$81.81
Implied value of everything else the parent owned	**–$60.78**

The market priced a profitable business, plus its cash, at minus $61 a share — and the stub stayed negative for 47 trading days. (3Com and Palm.)

Why nobody fixed it

The trade is obvious: short the subsidiary, buy the parent. But the subsidiary's shares could barely be borrowed.

2.6% → 47%

short interest, within one month of the IPO

≈ 40% a year

cost of borrowing the shares to short them

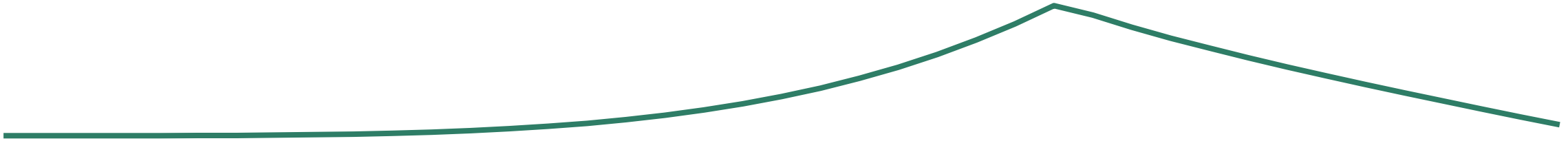
This is implementation costs, happening in public.

Closed-end funds make the same point quietly: a fund with a fixed number of shares trades on an exchange while holding listed securities you can price exactly — and its shares routinely trade 10–20% below what they hold.

Bubbles and Behavioral Economics

9.1

Easy to identify afterwards, genuinely hard to act on at the time.



1

Recent gains

Real, visible, and easy to project forward.

2

A new era

Representativeness — a few good years become a permanent condition.

3

Doubt driven out

Overconfidence suppresses it; limits to arbitrage drive out those who are right too early.

4

Collapse

The implied growth was never attainable.

Standard examples: technology stocks in the late 1990s; U.S. housing in the 2000s.

Why you cannot call one in advance. A high price is a bubble only if the growth it implies is impossible — and that takes a long-run growth forecast, which is exactly what nobody can make reliably. So “that was a bubble” is almost always said in the past tense.

Evaluating the Behavioral Critique

9.1

In its favour

- The biases are real and documented outside finance, in controlled experiments.
- Limits to arbitrage are a genuine mechanism, and the law-of-one-price cases show mispricing surviving in plain sight.
- It explains why the patterns of Chapter 8 might be stable rather than accidental.

Against

- There is a bias available to explain almost any pattern, and the choice is usually made after the pattern is known.
- Over- and underreaction appear in the record roughly equally often, which is what chance would also produce.
- It offers no single model that says what prices should be — only reasons they might be wrong.

A fair summary: behavioral finance has been more successful at explaining why prices can be wrong than at saying what they should be. That is a real contribution and a real limitation at the same time — and it is why the field has not displaced the rational framework so much as qualified it.

9.2

Technical Analysis and Behavioral Finance

- Why the two are connected
- Trends, moving averages, and breadth
- Machine learning applied to price patterns
- Sentiment indicators — and a warning

Why Behavioral Finance Revives Technical Analysis

9.2

If prices absorb news slowly — because investors are conservative — then the adjustment takes time.
And a slow adjustment is a trend.

Chapter 8 said no

Weak-form efficiency leaves no pattern in past prices to trade on.

Chapter 9 says maybe

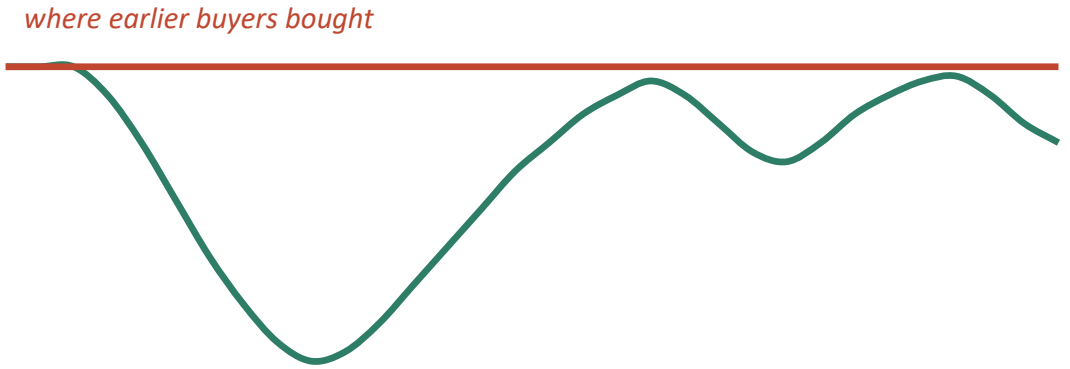
The mistakes that create a pattern persist, and arbitrage against them is limited.

Coherent, not correct

A reason a pattern could exist is not evidence that any rule finds it — and the rule must still beat its costs.

A mechanism you can draw: the resistance level

Investors hate selling at a loss. So as the price climbs back to where earlier buyers bought, those buyers sell to break even — and the price stalls. That stall is the chartist's “resistance level.”



Schematic. The stall is not in the chart — it is in the investors who bought at that price.

Trends and Corrections: Dow Theory

9.2

Three movements inside one price series

Dow's own image: a price series is a sea. Three movements are happening at once — and only one of them matters.

Tide	**Primary trend**	Years. The only movement an investor should care about.
Wave	**Secondary correction**	Weeks to months. Runs against the tide, then comes back.
Ripple	**Daily fluctuation**	Noise.

Standing on the beach, one wave tells you nothing about whether the tide is coming in or going out.

How you know the tide has turned

While it is rising — higher highs and higher lows:

100 → **120** → 110 → **135** → 125 → **150**

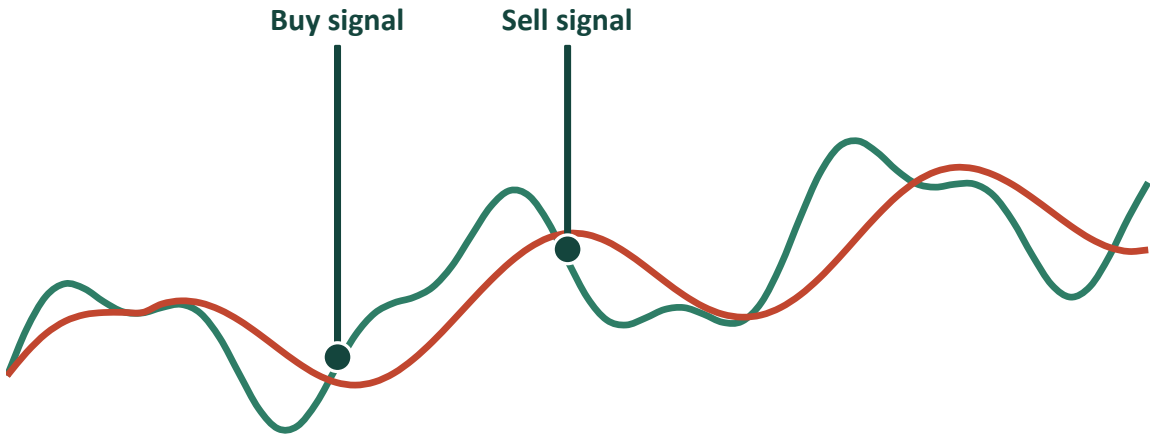
The turn needs two events. One alone is just a wave.

1	150 → 140 → **148**	A rally fails to exceed the previous high of 150.
2	148 → **135**	The next decline breaks the previous low of 140.

Trends and Corrections: Moving Averages

An average of the past, drawn against the present

Today's 50-day moving average is the mean of the last 50 closing prices. **Tomorrow one price drops out of the window and one comes in.**



***Price** against its **moving average**. Every crossing of the two lines is a signal.*

The signal

Price crossing above its moving average is read as bullish; crossing below, bearish. The chart marks one of each.

Watch one window slide

Daily closes: **50 52 51 55 57 54 58**

Day	Price	Five-day average
5	57	(50 + 52 + 51 + 55 + 57) / 5 = 53.0
6	**54**	(52 + 51 + 55 + 57 + 54) / 5 = **53.8**
7	58	(51 + 55 + 57 + 54 + 58) / 5 = 55.0

Day 6: the price fell from 57 to 54 — and the average still **rose**, 53.0 to 53.8. That is the lag, in one row.

Trends and Corrections: Breadth

9.2

How many issues are actually participating

The other three tools ask where the price is going. **Breadth asks how much of the market is behind the move.**

How an index can rise while most stocks fall

A cap-weighted index of ten stocks, on one day:

	Weight	Move	Contribution
One large stock	55%	+20%	+11.00%
Nine small stocks	45%	−3% each	−1.35%
The index	**100%**		**+9.65%**

Advances − declines = 1 − 9 = **−8**

The index is up almost 10%. Nine stocks out of ten went down.

The signal

When the index makes a new high but the advance–decline line does not, the rally is narrow, and is read as weak.

The advance–decline line, over four days

Each day's net is added to a running total:

Day	Index	Advances − declines	Running total
1	100.0	+120	+120
2	101.2	−40	+80
3	102.1	−90	−10
4	103.4	−150	−160

The index rises every day. The running total turns down and goes negative. The rally is being carried by fewer and fewer issues.

Trends and Corrections: Relative Strength

9.2

One price divided by another

The ratio of a stock's price to an index. **Rising is not the same as outperforming — and only the second one is what this measures.**

The same stock, over four periods

The ratio is scaled by 100 so it reads easily. Only its direction matters.

Period	Stock	Index	Ratio × 100
1	40.0	2,000	2.00
2	**44.0 (+10%)**	**2,300 (+15%)**	**1.91**
3	48.0 (+9%)	2,400 (+4%)	2.00
4	50.0 (+4%)	2,350 (−2%)	2.13

Period 2: the stock rose 10% — and its relative strength fell.
The index rose faster, so the stock lost ground while making money.

What the rule assumes

That the direction of the ratio persists — a stock beating the index will keep beating it. That is a momentum claim, and it is testable.

What the evidence shows

Over 3–12 months, largely yes: past winners keep beating past losers (momentum, as in Chapter 8). Over 3–5 years the pattern reverses, and trading costs absorb much of the gain.

Sentiment Indicators: Trin Statistic

9.2

Sentiment indicators · 1 of 3 — where the volume went, not where the prices went

More issues can rise than fall while most of the money moves the other way. **Trin is the measure of that.**

$$Trin = \frac{\frac{Volume\ declining}{Number\ declining}}{\frac{Volume\ advancing}{Number\ advancing}}$$

One day on the exchange

	Issues	Share Volume (m)	Average
Advancing	1,200	900	0.75
Declining	800	1,000	1.25

Trin = 1.25 / 0.75 = 1.67 → above 1: bearish

More issues rose than fell — and the average faller traded far more heavily.

How it is read

Above 1 is bearish: the average declining issue traded more heavily than the average advancing one, so the money is said to be leaving.
Below 1 is bullish.

Sentiment Indicators: Confidence Index

9.2

Sentiment indicators · 2 of 3 — a ratio of two yields, read as the market's nerve

Top-rated bonds always yield less than intermediate-grade ones, so the index is always below 1.
What is supposed to matter is which way it moves.

$$Confidence\ index = \frac{Yield\ on\ top-rated\ bonds}{Yield\ on\ intermediate-grade\ bonds}$$

What it is meant to show

	Top	Mid	Spread	Index
Before	5.0%	7.0%	2.0	0.714
After	5.0%	6.0%	1.0	0.833

The spread narrows and the index rises. Read as growing confidence.

But it is a ratio, not a spread

	Top	Mid	Spread	Index
Before	5.0%	7.0%	2.0	0.714
After	**6.0%**	**8.0%**	**2.0**	**0.750**

Both yields up one point. The spread is unchanged — and the index still rises.

The catch. A rise in the index is routinely described as a narrowing spread. It is not the same thing. The index also rises whenever both yields rise together, which is a story about the level of interest rates, not about confidence in credit.

Sentiment Indicators: Put/Call Ratio

9.2

Sentiment indicators · 3 of 3 — the same number, two opposite conclusions

Puts are bets on a fall, calls are bets on a rise. **A rising ratio means the crowd is turning bearish.**

$$Put/call\ ratio = \frac{Outstanding\ put\ options}{Outstanding\ call\ options}$$

Open interest, two months

	Puts (m)	Calls (m)	Ratio
Last month	1.2	2.0	0.60
This month	1.8	2.0	0.90

Half again as many puts outstanding, against the same number of calls.

The reading is 0.90. **Now what?**

Follow the crowd

Investors are bearish, and they may know something.
Expect the market to fall.

Fade the crowd

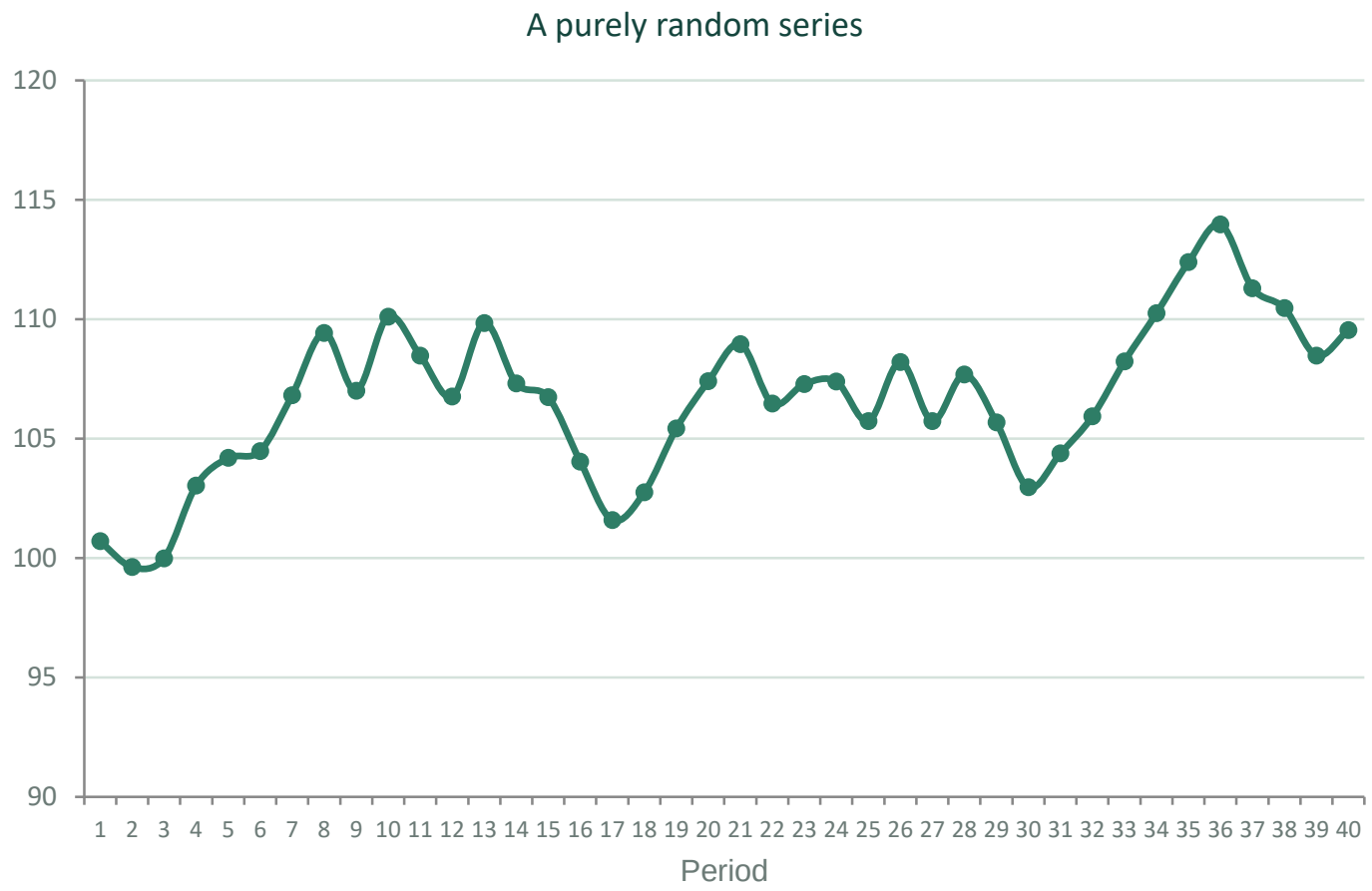
Investors are too bearish. The pessimism is already in the price.
Expect a rise.

Same number. Opposite trades.

The indicator does not tell you which reading to take. A rule that can be read either way cannot be wrong — and therefore cannot be tested.

A Warning: Patterns in Random Data

Schematic — this series was generated by coin flips.



The point

There is no trend here, no support, no resistance and no information of any kind. Every movement is a coin flip.

And yet the eye finds tops, bottoms and channels in it, because the eye finds those things in any series.

This is why a technical rule has to be tested the way any other hypothesis is tested, on data that was not used to find it. Looking convincing is not evidence — random data looks convincing too.